Linear Algebra II HW 1

Due Date: on or before 18:00, Feb/20/2024 (Thursday), in tutorial.

1. Prove that there exists a linear transformation $T: R^2 \to R^3$ such that $T(1,1) = (1,0,2)$ and $T(2,3) = (1,-1,4)$. What is $T(8,11)$?

2. Is there a linear transformation $T: R^3 \to R^2$ such that $T(1,0,3) = (1,1)$ and $T(-2,0,-6) = (2,1)$?

3. Determine whether the following are linear transformations from $\mathbb{R}^2$ into $\mathbb{R}^2$:

(a) $T(a_1, a_2) = (1, a_2)$
(b) $T(a_1, a_2) = (a_1, a_1^2)$
(c) $T(a_1, a_2) = (\sin a_1, 0)$
(d) $T(a_1, a_2) = (a_1, a_2, a_1 + 2a_2)$
(e) $T(a_1, a_2) = (a_1 + 1, a_2)$

4. Determine whether the following are linear transformations from P_2 to P_3:

(a) $L(p(x)) = xp(x)$
(b) $L(p(x)) = x^2 + p(x)$
(c) $L(p(x)) = p(x) + xp(x) + x^2 p'(x)$

5. Find the kernel and range of each of the following linear operators on P_3:

(a) $L(p(x)) = xp'(x)$
(b) $L(p(x)) = p(x) - p'(x)$
(c) $L(p(x)) = p(0)x + p(1)$

6. Find the *range* and *kernel* (those are new words for the column space and nullspace) of T.

(a) $T(v_1, v_2) = (v_2, v_1)$. (b) $T(v_1, v_2, v_3) = (v_1, v_2)$.
(c) $T(v_1, v_2) = (0, 0)$. (d) $T(v_1, v_2) = (v_1, v_1)$.

7. Let X and Y be 2-by-2 matrices such that

$$Tr(X)Y + Tr(Y)X = \begin{pmatrix} 4 & 8 \\ 4 & -4 \end{pmatrix} \quad \text{and} \quad XY = \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix}.$$

Find all possible solution for X and Y.

8. Let A be a m-by-n matrix and B be a n-by-m matrix. Show that $Tr(AB) = Tr(BA)$. If $m = n$, is it possible to find 2 m-by-m matrices A and B such that $AB - BA = I_m$.